

KRUX Finance - Tech Intern Assessment

📄 Overview

Build a customer support system that demonstrates your ability to work with the same technologies we use in our production application.

🎯 Take-Home Assignment: Dual Chatbot System

Project Requirements

Build a comprehensive customer support system with **two distinct interfaces**:

1. **Customer Chatbot** - For end users seeking help with loans
2. **Support Executive Dashboard** - For customer support agents to manage conversations

Technical Stack Requirements

Use the **same technologies as KRUX Finance v2**:

- **Next.js 14+** with **TypeScript** (App Router)
- **Tailwind CSS** for styling
- **React Hook Form** with **Zod** validation
- **Lucide React** icons
- **Context API** for state management
- Any **component library** of your choice
- **Local storage** for data persistence (no backend required)
- **Mock authentication** (sample users)

Core Features Overview

Page 1: Customer Chatbot Interface

Customer Chat Experience

What customers should be able to do:

- Start a conversation about loan applications
- Ask questions about document requirements
- Check their application status
- Request to speak with a human agent
- Get helpful responses from an automated bot

UI/UX Requirements:

- Mobile-first design (like WhatsApp/Telegram)
- Clean, professional interface matching KRUX Finance branding
- Typing indicators and message timestamps
- Smooth scrolling and responsive design

Bot Conversation Flows

****The bot should handle these scenarios:****

****Loan Application Help:****

- Guide users through loan application process
- Explain different loan types (Business, Personal, MSME)
- Provide application requirements and next steps

****Document Requirements:****

- List required documents for different loan types
- Explain document formats and specifications
- Help with document-related queries

****Application Status:****

- Allow users to check their application status
- Provide realistic status updates and timelines
- Handle cases where application ID is not found

****Escalation to Human Agent:****

- Seamlessly transfer users to human support
- Create support tickets for agent follow-up
- Maintain conversation context during transfer

Page 2: Support Executive Dashboard

Support Agent Experience

****What support agents should be able to do:****

- View all customer support tickets in a queue
- See ticket priorities and categories
- Chat with customers in real-time
- Access customer information and loan history
- Use quick reply templates
- Resolve, escalate, or transfer tickets
- Track response times and performance

Dashboard Layout

****Suggested interface structure:****

- ****Left Panel****: Ticket queue with priority indicators
- ****Right Panel****: Active conversation with customer
- ****Top Bar****: Agent status, notifications, search
- ****Bottom****: Message input with agent tools

Agent Tools & Features

****Essential agent functionality:****

- ****Quick Reply Templates****: Pre-written responses for common queries
- ****Customer Information Panel****: Display customer's loan applications and status
- ****Ticket Management****: Ability to resolve, escalate, or transfer conversations
- ****Internal Notes****: Private notes not visible to customers
- ****Response Time Tracking****: Monitor how long customers wait for responses
- ****Real-time Synchronization****: Updates between customer chat and agent dashboard

Application Structure

****Create a Next.js application with these pages:****

- ****Landing Page****: Navigation between customer chat and support dashboard
- ****Customer Chat Page****: `/customer-chat`` - Public access for customers
- ****Support Dashboard****: `/support-dashboard`` - Protected access for agents

Authentication System

****Implement simple mock authentication:****

Data Management

****Use localStorage to store:****

- Chat conversations and message history
- Support tickets and their status
- User sessions and authentication state
- Agent notes and customer information

State Management

****Use React Context API and useReducer for:****

- Managing chat messages and conversation state
- Handling support ticket queue and updates
- Synchronizing data between customer and agent interfaces
- Managing authentication and user sessions

Bonus Features (Extra Credit)

****Implement any of these for additional points:****

1. ****Quick Reply Templates**** - Pre-defined responses for agents
2. ****File Upload Simulation**** - Mock document upload functionality
3. ****Chat History Persistence**** - Save and restore conversation history
4. ****Real-time Simulation**** - Use polling or intervals for live updates
5. ****Dark/Light Mode**** - Theme switching for both interfaces
6. ****Voice Input**** - Speech-to-text using browser APIs
7. ****Performance Metrics**** - Track agent response times

8. ****Customer Satisfaction**** - Post-chat rating system
9. ****Search Functionality**** - Search through tickets and conversations
10. ****Notification System**** - Visual/audio alerts for new messages

Sample Login Credentials

****Customer:****

- Phone: +919876543210 (Rahul Sharma)
- Phone: +919876543211 (Priya Patel)

****Agent:****

- Username: amit.kumar (Support Agent)
- Username: sneha.singh (Senior Agent)

Features Implemented

- [x] Customer chat interface
- [x] Support dashboard interface
- [x] Bot conversation flows
- [x] Mock authentication
- [x] Local storage data persistence
- [] Voice input (bonus)

3. Live Demo

- Deploy to ****any hosting platform**** (Vercel, Netlify, GitHub Pages, etc.)
- Provide live demo URL
- Include demo credentials in your README
- Ensure both customer and agent interfaces work in production

Time Limit

****1-2 days**** from assignment delivery